

Technology Stack

Date	16 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API
Maximum Marks	2 Marks

Technology Stack Details

The AI FAQ Assistant is a headless RESTful backend; API tools (Postman / Thunder Client) consume the REST endpoints directly during development and grading, ahead of any dedicated frontend.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Postman / Thunder Client (VS Code extension) as the API client during this cycle	Lets the team verify every endpoint independently of any separate frontend build
2	Backend / Server-Side	Node.js + Express.js, with <code>cors</code> and <code>dotenv</code> -driven configuration	Non-blocking I/O suits the Gemini AI calls; Express keeps routing, middleware and controllers cleanly separated
3	Database / Data Storage	MongoDB with Mongoose ODM	Mongoose schema validation (required fields, enum, unique index) keeps FAQ and User documents consistent without hand-written checks in every controller
4	Cloud / Hosting / Deployment	API server run locally on port 5000 by default (<code>process.env.PORT 5000</code>) – the team's captured test session ran it on port 5001 via a local <code>.env</code> override – for the 15–29 September 2026 grading window	Fully <code>.env</code> -driven configuration keeps the same codebase portable to a managed host later
5	Version Control & CI/CD	Git & GitHub – github.com/sabarivasan-2105/AI_FAQ_Assistant_API	Single source of truth for the 5-member team; commit history traces each Epic
6	Third-Party APIs / Other Tools	Google Gemini AI (<code>@google/genai</code>) for direct-answer and FAQ-generation flows; <code>jsonwebtoken</code> + <code>bcrypt</code> (auth/security); <code>dotenv</code> , <code>cors</code> (config/CORS); Postman collection for endpoint verification	Each library enables one specific, named feature of the solution — generative FAQ content, secure auth, and config isolation